

SALES DATA CLEANING AND PREPROCESSING USING PYTHON

Project Report

Submitted by:

Thamindu Kavinda

Data Analytics Intern DecodeLabs

Submitted To:

DecodeLabs

Submission Date:

2026/05/19

Declaration

I certify that the project report titled "Data Cleaning and Preprocessing for Sales Data Analysis using Python" was written by me as a requirement for the Data Analytics Internship Program at DecodeLabs.

The content included in this report is entirely original except where cited.

Signature: __Thamindu Kavinda__

Name: Thamindu Kavinda

Date: ____2026/05/19____

Acknowledgement

I am honored to express my sincere gratitude to DecodeLabs for giving me the opportunity to complete this Data Analytics Internship Project.

Thank you to all the mentors and advisors who guided me during this project.

This project gave me hands-on experience in data cleaning and preprocessing using Python and Pandas.

Table of Contents

1. Introduction
2. Objectives
3. Technologies Used
4. Dataset Description
5. Methodology
6. Results and Outputs
7. Conclusion
8. Future Improvements
9. References

1. Introduction

The aim of the current project is to prepare the dataset for analysis by performing various data cleaning processes. Data cleaning is an essential part of the data analysis process as raw data usually has many issues such as missing values, duplicates and other inconsistencies.

The main goal of the current project is to prepare the data before further data analysis is performed. A number of data preprocessing processes are applied.

2. Objectives

- To load and inspect the dataset
- To identify missing values
- To detect duplicate records
- To clean inconsistent data
- To standardize date formats
- To create a cleaned dataset for analysis

3. Technologies Used

- Python
- Pandas
- Google Colab
- Microsoft Excel

4. Dataset Description

The dataset used in the project provided data on sales transactions. The dataset includes details such as order ID, date, coupon codes, and customer details.

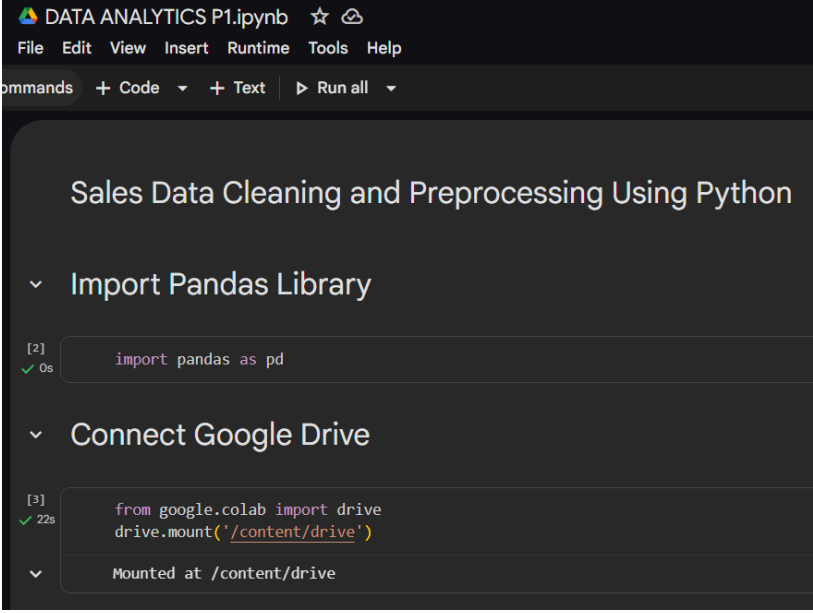
The above dataset was used for data cleaning and preprocessing.

5. Methodology

- Importing Libraries
- Loading Dataset
- Dataset Inspection
- Missing Value Detection
- Duplicate Detection
- Removing Duplicates
- Date Conversion
- Filling Missing Values
- Exporting Cleaned Dataset

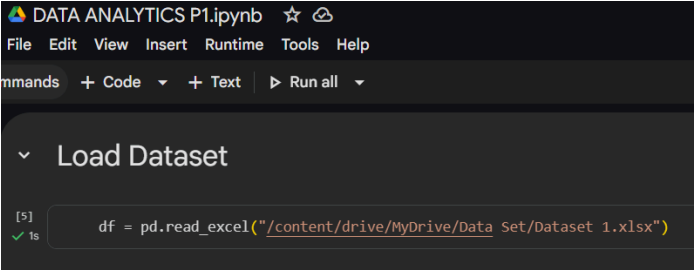
6. Project Screenshot

- Import Library and Connect google drive

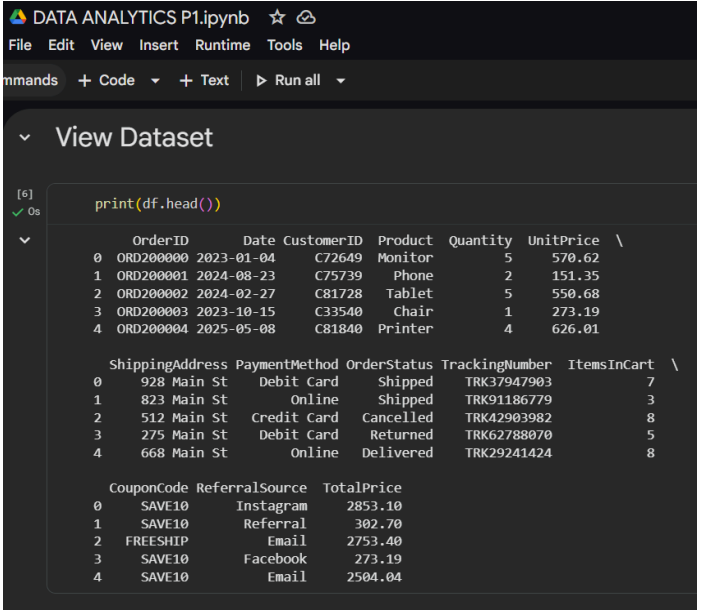


```
DATA ANALYTICS P1.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Sales Data Cleaning and Preprocessing Using Python
Import Pandas Library
[2] import pandas as pd
Connect Google Drive
[3] from google.colab import drive
drive.mount('/content/drive')
Mounted at /content/drive
```

- Loading the Dataset and Dataset Inspection



```
DATA ANALYTICS P1.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Load Dataset
[5] df = pd.read_excel('/content/drive/MyDrive/Data Set/Dataset 1.xlsx')
```



```
DATA ANALYTICS P1.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
View Dataset
[6] print(df.head())
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	
0	928 Main St	Debit Card	Shipped	TRK37947903	7	
1	823 Main St	Online	Shipped	TRK91186779	3	
2	512 Main St	Credit Card	Cancelled	TRK42903982	8	
3	275 Main St	Debit Card	Returned	TRK62788070	5	
4	668 Main St	Online	Delivered	TRK29241424	8	

	CouponCode	ReferralSource	TotalPrice
0	SAVE10	Instagram	2853.10
1	SAVE10	Referral	302.70
2	FREESHIP	Email	2753.40
3	SAVE10	Facebook	273.19
4	SAVE10	Email	2504.04

- Checking Dataset Information

```

[7]
✓ Os
print(df.info())

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   OrderID               1200 non-null   object  
 1   Date                  1200 non-null   datetime64[ns]
 2   CustomerID            1200 non-null   object  
 3   Product               1200 non-null   object  
 4   Quantity              1200 non-null   int64   
 5   UnitPrice             1200 non-null   float64  
 6   ShippingAddress        1200 non-null   object  
 7   PaymentMethod          1200 non-null   object  
 8   OrderStatus           1200 non-null   object  
 9   TrackingNumber         1200 non-null   object  
10   ItemsInCart           1200 non-null   int64   
11   CouponCode            891 non-null    object  
12   ReferralSource         1200 non-null   object  
13   TotalPrice            1200 non-null   float64  
dtypes: datetime64[ns](1), float64(2), int64(2), object(9)
memory usage: 131.4+ KB
None

```

- Missing Value Detection

```

[8]
✓ Os
print(df.isnull().sum())

OrderID      0
Date         0
CustomerID   0
Product      0
Quantity     0
UnitPrice    0
ShippingAddress 0
PaymentMethod 0
OrderStatus  0
TrackingNumber 0
ItemsInCart  0
CouponCode   309
ReferralSource 0
TotalPrice   0
dtype: int64

```

- Duplicate Record Detection and Removing Duplicate Records

The screenshot shows a Jupyter Notebook titled "DATA ANALYTICS P1.ipynb" with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar (+ Code, + Text, Run all). The notebook content is organized into three sections:

- Duplicate Rows Check**: Cell [9] contains `print(df.duplicated().sum())`, which outputs `0`.
- Duplicate IDs Check**: Cell [10] contains `print(df["OrderID"].duplicated().sum())`, which outputs `0`.
- Remove Duplicates**: Cell [11] contains `df = df.drop_duplicates()`.

- Date Format Conversion

The screenshot shows a Jupyter Notebook titled "DATA ANALYTICS P1.ipynb" with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar (+ Code, + Text, Run all). The notebook content includes a section titled "Convert Date Format" and a code cell [20] with the following code:

```
df["Date"] = pd.to_datetime(df["Date"])
```

The screenshot shows a Jupyter Notebook titled "DATA ANALYTICS P1.ipynb" with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar (+ Code, + Text, Run all). The notebook content includes a section titled "Check Date Format" and a code cell [18] with the following code:

```
print(clean_df["Date"].head())
```

The output of the code is a table showing the first five rows of the "Date" column:

	Date
0	2023-01-04
1	2024-08-23
2	2024-02-27
3	2023-10-15
4	2025-05-08

Below the table, the output indicates: `Name: Date, dtype: datetime64[ns]`.

- Handling Missing Values

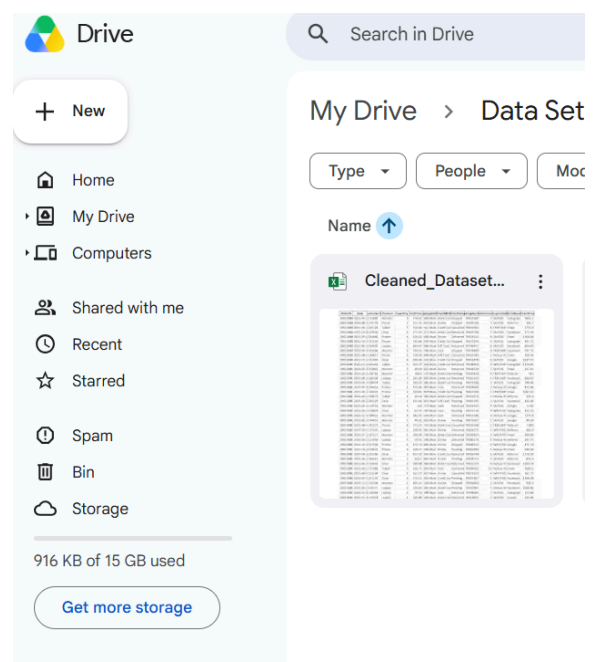
```
DATA ANALYTICS P1.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Fill Missing Values
[13] df["CouponCode"] = df["CouponCode"].fillna("Notuse the Coupon")
✓ 0s
```

- Verify Dataset

```
DATA ANALYTICS P1.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Verify Cleaning
Load Clean Dataset
[15] clean_df = pd.read_excel("/content/drive/MyDrive/Data Set/Cleaned_Dataset.xlsx")
✓ 0s
Check Missing
[16] print(clean_df.isnull().sum())
✓ 0s
OrderID      0
Date         0
CustomerID   0
Product      0
Quantity     0
UnitPrice    0
ShippingAddress 0
PaymentMethod 0
OrderStatus  0
TrackingNumber 0
ItemsInCart  0
CouponCode   0
ReferralSource 0
TotalPrice   0
dtype: int64
Check Duplicate Rows and IDs
[17] print("Duplicate Rows=", clean_df.duplicated().sum())
print("Duplicate IDs=", clean_df["OrderID"].duplicated().sum())
✓ 0s
Duplicate Rows= 0
Duplicate IDs= 0
```

- Exporting the Cleaned Dataset

```
DATA ANALYTICS P1.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Save Cleaned Dataset
[14] df.to_excel("/content/drive/MyDrive/Data Set/Cleaned_Dataset.xlsx", index=False)
✓ 0s
```



- Final Cleaned Dataset

Final Dataset Preview

```
[19] print(clean_df.head())
```

✓ Os

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	\
0	928 Main St	Debit Card	Shipped	TRK37947903	7	
1	823 Main St	Online	Shipped	TRK91186779	3	
2	512 Main St	Credit Card	Cancelled	TRK42903982	8	
3	275 Main St	Debit Card	Returned	TRK62788070	5	
4	668 Main St	Online	Delivered	TRK29241424	8	

	CouponCode	ReferralSource	TotalPrice
0	SAVE10	Instagram	2853.10
1	SAVE10	Referral	302.70
2	FREESHIP	Email	2753.40
3	SAVE10	Facebook	273.19
4	SAVE10	Email	2504.04

7. Conclusion

The project emphasized the importance of data cleaning in the data analysis process. Various methods of data preprocessing were used using the programming language Python and its library Pandas.